

URL SHORTENER APPLICATION

A **URL Shortener** is a web application that converts a long URL (Uniform Resource Locator) into a shorter and more manageable link. This helps users share links easily without copying long web addresses.

URL shorteners work by creating a **redirect** from the short URL to the original long URL. When the short URL is opened in a browser, the user is redirected automatically to the original website.

Project Overview

This project is an advanced version of URL Shortener Web Application built using Flask (Python) as backend, HTML/CSS/Bootstrap as frontend, and SQLite as database using SQLAlchemy ORM. The application allows users to create accounts, login securely, shorten long URLs, copy shortened URLs, and manage their URL history with pagination and search.

Objectives

The main objectives of this application are:

- Shorten long URLs into compact and shareable short links.
- Allow users to save URLs by logging into their account.
- Provide history management with delete and clear-all options.
- Provide a clean, responsive, fit-to-display UI.
- Display saved URLs in a tabular format with pagination (10 per page).

The project consists of 2 parts:

- Frontend: HTML5, CSS3, Bootstrap 5
- Backend: Flask (Python)
- Database: SQLite
- ORM: SQLAlchemy
- Authentication: Flask-Login + Werkzeug Password Hashing
- URL Shortening Service: TinyURL (via pyshorteners)

Front-End Information

The front-end consists of 4 pages in this implementation:

- 1) Home Page - Provides Login and Signup options.
- 2) Signup Page - User enters username and password.
- 3) Login Page - User logs into the system.
- 4) Dashboard Page - URL shortening + history management.

Project Workflow

Signup Workflow:

- User enters a username and password.
- Username must be unique.
- Username length must be between 5 and 9 characters.
- On success, user is redirected to Login page.

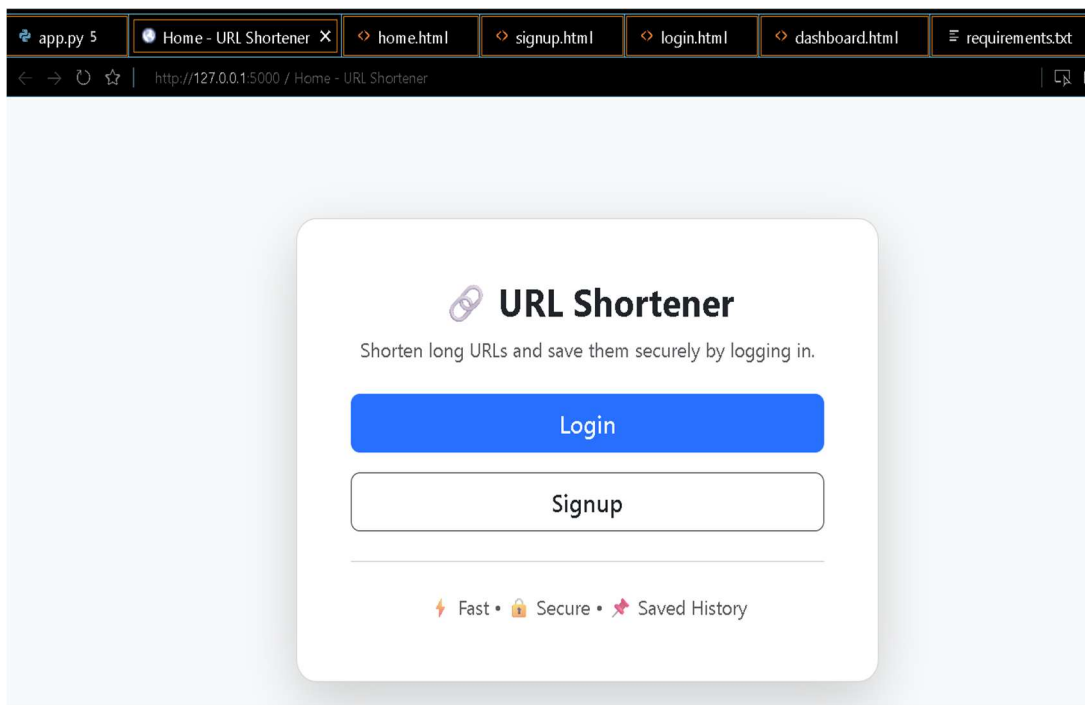
Login Workflow:

- User logs in with username and password.
- On success, user is redirected to Dashboard.

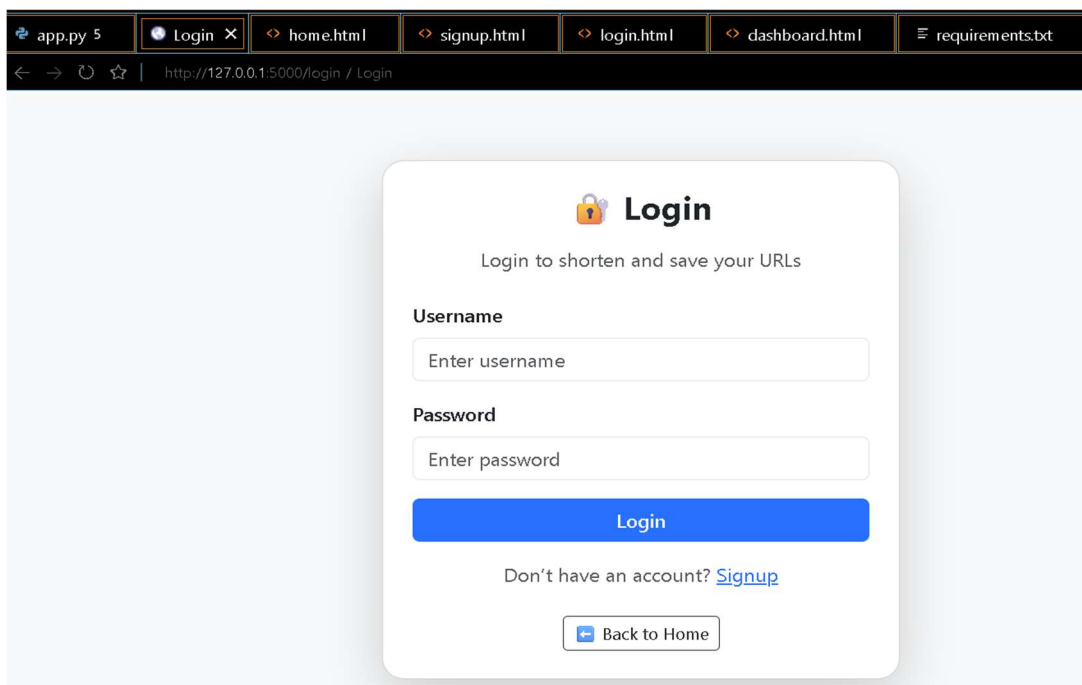
URL Shortening Workflow:

- User enters a valid URL (http/https).
- Application generates a shortened URL using TinyURL.
- The URL pair is stored in SQLite database linked to the user.
- History can be viewed by clicking 'Show History'.

Page 1 (Home Page)



Page 2a (Login Page)



Page 2b (Signup Page)

app.py 5 Signup X home.html signup.html login.html dashboard.html requirements.txt

http://127.0.0.1:5000/signup / Signup

Signup

Create your account (5-9 characters)

Username

Password

Create Account

Already have an account? [Login](#)

[Back to Home](#)

Page 3 (Dashboard)

app.py 5 Dashboard X home.html signup.html login.html dashboard.html requirements.txt

http://127.0.0.1:5000/dashboard / Dashboard

URL Shortener

Logout

Welcome, admin 🙌

Shorten your URLs and manage your history.

Login successful! X

Enter Long URL

Shorten URL

[Show History](#) [Clear All](#)

app.py 5

Dashboard X

home.html

signup.html

login.html

dashboard.html

requirements.txt

http://127.0.0.1:5000/dashboard / Dashboard

URL Shortener

Logout

Welcome, admin 🙌

Shorten your URLs and manage your history.

URL shortened successfully! X

Enter Long URL

Shorten URL

Shortened URL

Copy

Show History

Clear All

app.py 5

Dashboard X

home.html

signup.html

login.html

dashboard.html

requirements.txt

http://127.0.0.1:5000/dashboard / Dashboard

URL Shortener

Logout

Welcome, admin 🙌

Shorten your URLs and manage your history.

Enter Long URL

Shorten URL

Hide History

Clear All

Your Previous URLs

#	Original URL	Short URL	Copy	Delete
1	https://www.msn.com/en-in/money/topstories/rain-forecast-for-north-india-raises-red-flags-here-s-why-it-s-concerning/ar-A1UunjA?cvid=907f30c8e4134521a72ca5f1251a2ce4&ei=4	https://tinyurl.com/23hzcjr3	Copy	Delete

Page 6 (Delete the URL)

 **Your Previous URLs**

#	Original URL	Short URL	Copy	Delete
1	https://www.msn.com/en-in/autos/news/6-k-dramas-starring-kim-seon-ho-go-youn-jung-to-watch-on-netflix-if-you-loved-the-hem-in-can-this-love-be-translated/ar-AA1UvzoY?cvid=3fc97d85b7b74b86f3dd80b3fe2bde29&ei=7	https://tinyurl.com/28awaxkj	<button>Copy</button>	<button>Delete</button>
2	https://www.msn.com/en-in/money/topstories/rain-forecast-for-north-india-raises-red-flags-here-s-why-it-s-concerning/ar-AA1UunjA?cvid=907f30c8e4134521a72ca5f1251a2ce4&ei=4	https://tinyurl.com/23hzcjr3	<button>Copy</button>	<button>Delete</button>
3	https://www.msn.com/en-in/news/india/cops-arrest-accused-driver-in-jaipur-audi-hit-and-run-case/ar-AA1UtTs?cvid=68c8584bacb94de4ed77b9d142741323&ei=28	https://tinyurl.com/29jmnae2	<button>Copy</button>	<button>Delete</button>

Page 7

app.py 5

Dashboard X

home.html

signup.html

login.html

dashboard.html

requirements.txt

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http://127.0.0.1:5000/dashboard / Dashboard

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 **URL Shortener**

Logout

Welcome, admin 🙌

Shorten your URLs and manage your history.

URL deleted successfully!

X

Enter Long URL

Shorten URL

 Show History

 Clear All

Page 8 (Logout)

app.py 5 Login X home.html signup.html login.html dashboard.html requirements.txt

http://127.0.0.1:5000/login / Login

Login

Login to shorten and save your URLs

Logged out successfully! X

Username

Password

Login

Don't have an account? [Signup](#)

[Back to Home](#)

This Advanced URL Shortener web application successfully demonstrates user authentication, database storage, URL shortening, and responsive UI design. It is a practical project to learn Flask, SQLAlchemy, Flask-Login, and Bootstrap based web development.

It allows users to shorten URLs, copy them easily, and store them for future use in the History page. The application is simple, responsive, and useful for everyday link sharing.